HEATMAP NODE · USER REQUIREMENTS

HeatMap Node user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First draft at the kit 1.9.1 upgrade, written from the users and stakeholders in HMN-PRB-001 v0.7 and used to score the concept options (HMN-OPT-001); each need traced to product requirements in HMN-PRS-001; to review with a co-design partner

HeatMap Node user requirements

What the people who will use, host and pay for a HeatMap network need, written from their side. The fourteen needs below come from the users and stakeholders in HMN-PRB-001 (Tables 1 and 2) and from the 2026-09-25 requirements work. They score the concept options (HMN-OPT-001) and are met by the product requirements in HMN-PRS-001 (links in HMN-RTM-001). They are a desk draft until a co-design partner, the first candidate being the City of Phoenix Office of Heat Response and Mitigation (HMN-DEC-001), reviews them with the people who would use the data and fit the nodes. Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Street-level heat stress	The node reports the heat a person feels on the street (heat stress, not only air temperature), with the air temperature, humidity and wind that explain it	High	City heat officer, public health department (PRB Table 1)
UR-002	Performance	Numbers I can trust	The readings are accurate enough to compare blocks and to stand beside a reference instrument, and the error is stated	High	Researcher, public health department
UR-003	Deployment and assembly	Fits the poles we have	Fits existing street light, sign or utility poles without drilling or welding	High	City heat officer, pole owner
UR-004	Deployment and assembly	Quick to fit	Two people fit a node from a ladder or lift in about half an hour	Medium	Pole owner, community group
UR-005	Logistics	Runs without mains	The node runs from its own solar panel through several days without sun	High	City heat officer, pole owner
UR-006	Performance	Near-real-time data	Data reaches the gateway within about half an hour, so a heat wave can be followed	High	Public health department
UR-007	Cost	Cheap enough for a network	The sensor head costs little enough that tens of nodes per neighborhood are possible	High	City heat officer, community group
UR-008	Environment and durability	Survives the street	Works through -20 to +50 °C at the node, full sun, driving rain, dust and wind gusts	High	City heat officer, pole owner
UR-009	Regulatory	Environmental data only	No camera, microphone or device tracking; nothing that identifies a person	High	Community group, residents
UR-010	Usability	Open, documented data	Readings come with the node's location, height and calibration record in an open, documented format	Medium	Researcher, community group
UR-011	Maintenance	Little upkeep	A visit is needed rarely and takes little skill or consumables	Medium	Pole owner, community group
UR-012	Safety	Safe overhead	Nothing falls on passers-by in wind, the node adds little load to the pole, and edges are safe	Medium	Pole owner, residents
UR-013	Performance	Measured where people walk	The sensors sit at or near pedestrian height	Medium	Public health department, researcher
UR-014	Logistics	Buildable by a community	A community group or workshop can build it from open files, common parts and a printer	Medium	Community group

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.